Churn Prediction and Intervention in Underserved Educational Communities using Restless Multi-Armed Bandits

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Abstract

Student attrition, or educational churn, remains a critical barrier to equity in online and blended learning environments, particularly within resource-constrained communities. While predictive analytics have advanced significantly, a gap remains in translating diagnostic risk scores into prescriptive, resource-efficient interventions. This paper proposes a novel, closed-loop framework: the State-Aware, Cost-Sensitive Bandit (SCSB). The system integrates a deep neural churn-prediction model (Slow Loop) with a Restless Multi-Armed Bandit (RMAB) intervention engine (Fast Loop). Unlike traditional policies that ignore resource expenditure, the SCSB explicitly calculates a "Cost-Adjusted Risk Reduction" utility score, optimizing for maximum engagement retention within a strict budget. We evaluate our framework on the Open University Learning Analytics Dataset (OULAD). Our results demonstrate that the proposed Bayesian Thompson Sampling policy significantly reduces cumulative regret compared to ϵ-greedy baselines and successfully converges to optimal intervention strategies while adhering to budgetary constraints. The system is further supported by an explainable teacher dashboard to ensure transparency and human-in-the-loop decision-making.

Keywords: Student Churn Prediction; Restless Multi-Armed Bandits; Educational Data Mining; Cost-Sensitive Learning; Thompson Sampling.

1 Introduction

1.1 Background

Student dropout, or educational churn, represents a persistent and systemic challenge across global educational institutions, particularly within the domain of online and blended learning and in underserved educational communities. The social and economic repercussions of attrition are substantial, impacting long-term student success, institutional funding, and societal equity. In response, the field of Educational Data Mining (EDM) has increasingly focused on developing predictive analytics to identify at-risk learners early in their academic lifecycle.

While early warning systems utilizing Machine Learning (ML) classifiers have demonstrated high accuracy in forecasting dropout events, a critical gap remains in translating these static predictions into effective, real-time, and resource-conscious prescriptive interventions. The deployment challenge is further amplified in low-resource environments, necessitating solutions that operate efficiently on minimal data while offering high interpretability for local educational staff.

The next generation of educational support systems must move beyond simple prediction toward a closed-loop framework where data not only identifies risks but dynamically selects the optimal action. This mandates leveraging advanced sequential decision-making models, such as those derived from the Multi-Armed Ban-

dit (MAB) framework, which are inherently designed to manage the classic dilemma between exploration (trying new interventions) and exploitation (relying on proven ones).

1.2 Scope of the Work

This research focuses on the intersection of student churn prediction and adaptive intervention scheduling within the challenging context of resource-constrained and underserved educational settings. The scope is specifically limited to designing an integrated, two-stage framework that transforms coarse-grained student activity into a quantifiable risk state and uses that state to select the most efficient, cost-effective intervention. By adopting the principles of the State-Dependent Multi-Armed Bandit with Abandonment (MAB-A), this work seeks to ensure that the adaptive strategy operates safely while explicitly minimizing the likelihood of student disengagement.

1.3 Problem Definition

The existing literature presents three major deficiencies that impede effective dropout prevention, particularly in low-resource contexts.

1. Firstly, most high-performing predictive models (e.g., those detailed in Krüger et al. [1]) are data-intensive and focus on static feature sets, often overlooking the rapid temporal fluctuations in student engagement that are critical for timely intervention.

2. Secondly, when intervention is considered, traditional bandit algorithms often fail because they assume a persistent user, neglecting the reality of abandonment risk [2]. This risk mandates conservative action when a student is already disengaged.

3. Finally, and most pertinent to underserved areas, adaptive systems often recommend high-cost actions (such as mentor calls) without factoring in the differential cost of resource allocation.

The challenge, therefore, is to create a dynamic system that simultaneously manages the technical regret of the learning algorithm, the pedagogical risk of student abandonment, and the practical constraint of intervention cost.

1.4 Technical Insights

To overcome the stated problem, the methodology must embed three core technical advancements:

State-Aware Adaptivity: The intervention logic must transition from standard sequential decision-making to a state-dependent policy. Inspired by the MAB-A model [2], this policy must adjust its exploration-exploitation balance based on the student's current engagement state (S). When the student is engaged ($S = 1$), the system should explore; when disengaged ($S = 0$), it must exploit proven, safer actions to mitigate the risk of abandonment.

Cost-Awareness: Existing models, even advanced bandit structures, typically optimize for reward or regret but ignore resource cost [2]. Our solution requires a novel modification to the bandit's value function or reward signal to explicitly penalize high-cost arms, thereby biasing the system towards low-cost interventions (e.g., automated SMS) unless high engagement risk justifies the expense.

Explainability and Trust: For the system to be adopted by teachers in low-resource settings, the black-box nature of the underlying models must be mitigated. The framework requires integrating an Explainable AI (XAI) component, aligning with Krüger et al. [1], to clearly present the top factors driving the dropout prediction to the teacher dashboard.

1.5 Objectives of the Paper

The objectives of the paper are:

- To conduct a thorough background study and enlighten the literature review with regard to student churn prediction and state-aware bandit-based intervention strategies in resource-constrained educational environments.

- To propose a novel, integrated methodology titled State-Aware, Cost-Sensitive Bandit (SCSB) Framework for solving the optimal, resource-efficient intervention problem. This technique improves upon the MAB-A benchmark by incorporating an explicit penalty for differential intervention cost.

- To test and validate the proposed methodology on key sequential performance metrics, primarily Cumulative Regret and Intervention Effectiveness (as measured by engagement improvement).

- To compare the proposed SCSB methodology with established benchmark techniques, including the base MAB-A model and traditional index policies (e.g., UCB and KL-UCB).

1.6 Research Questions (RQ)

The core research questions addressed by this work are:

- **RQ1:** How effectively does the proposed SCSB framework balance the Exploration-Exploitation-Engagement trade-off compared to non-cost-aware MAB-A models?

- **RQ2:** Can integrating an explicit cost-awareness mechanism into the bandit's policy significantly improve overall intervention efficiency (maximize reward while minimizing resource cost) in a simulated resource-constrained environment?

- **RQ3:** What is the minimal set of readily available data attributes (e.g., assignment timeliness, login frequency) required to establish a robust Engagement State capable of supporting the SCSB decision-making process?

1.7 Organization of the Paper

The rest of the paper is organized as follows: Section 2 presents the related work, organizing the literature around prediction, explanation, and bandit intervention strategies. Section 3 describes the Methodology, focusing on the dataset, prediction model, and intervention strategy. Section 4 details the proposed State-Aware, Cost-Sensitive Bandit (SCSB) Architecture and its mathematical formulation. Section 5 explains the experimental setup and simulation results. Section 6 concludes the paper and outlines future research directions.

2 Related Work

Chaikalis [3] [2024] proposed supervised machine learning (KNN, Naive Bayes) to predict student dropouts using diverse factors via the EDUC8EU platform. Its novelty was integrating varied attributes into a single model for early identification. Experimentally, Naive Bayes consistently outperformed KNN, achieving 80.46% accuracy (vs. 79.59%) and proving more stable for dropout prediction.

Psyridou et al. [4] [2024] proposed using machine learning (e.g., Balanced Random Forest) to predict upper secondary dropout, leveraging a novel 13-year longitudinal dataset from kindergarten. This enabled predictions as early as Grade 6. The Balanced Random Forest classifier performed best, achieving a mean AUC of 0.65 (Grade 9 data) and 0.61 (Grade 6 data).

Yang et al. [2] [2024] proposed MAB-A, a new Multi-Armed Bandit model addressing user abandonment based on engagement, and introduced the ULCB and KL-ULCB algorithms. Its novelty lies in a state-dependent exploration-exploitation-engagement tradeoff. Simulations showed that these algorithms achieved significantly lower regret than traditional UCB/KL-UCB and Q-learning, with KL-ULCB attaining asymptotically sharp regret bounds.

Xia and Qi [5] [2023] proposed DPM-MTS, a dropout prediction model for MOOCs, analyzing multi-temporal learning behavior sequences using parallel full-convolutional neural networks. Its novelty lies in integrating diverse temporal features and noise filtering to capture long- and short-term trends, surpassing single-interval methods. Experimentally, DPM-MTS outperformed benchmarks and deep learning models, achieving an AUC of 0.904.

Vaarma and Li [6] [2024] proposed using machine learning (CatBoost, NN, LR) to predict Finnish higher education dropouts, integrating LMS, transcript, and demographic data. Its novelty lay in the monthly analysis and comparison of data source importance. "Accumulated credits," "failed courses," and "Moodle activity count" were key predictors, with performance improving over time.

Berens et al. [7] [2025] proposed the use of machine learning to predict higher education dropouts nationally in Switzerland, utilizing administrative data to track stu-

dent trajectories across the entire system. Its novelty lay in this comprehensive national perspective, differentiating true dropouts from transfers. With pre-enrollment data, it achieved 73% accuracy (AUC 79), rising to 88% (AUC 89) after the fourth semester by incorporating academic performance data.

Tio et al. [8] [2024] proposed EduQate, an interdependency-aware Reinforcement Multi-armed Bandit model, for generating adaptive educational curricula. Its novelty lies in modelling interdependencies between learning content, which traditional RMABs neglect. Experimentally, EduQate outperformed baselines on synthetic and real-world datasets, demonstrating its effectiveness in maximizing student learning.

Cai et al. [9] [2021] developed MathBot, a rule-based chatbot, integrating a contextual bandit algorithm to personalize the pace of math instruction. Its novelty is the live deployment of a contextual bandit in a conversational educational system. Experimentally, the bandit learned an effective policy with lower experimentation cost (less time, reduced dropout) than A/B tests, achieving comparable learning gains.

Krüger et al. [1] [2023] used an end-to-end framework with Explainable Machine Learning (XGBoost + SHAP) to predict dropout and provide interpretable justifications via dashboards. Novelty included the explanatory focus and enriching the dataset with temporal features (e.g., cumulative grade). The XGBoost model achieved strong AUC-PR scores up to 89.5%.

2.1 Gap Identification

The comprehensive review of the literature, spanning prediction models, platform design, and specialized bandit theory, reveals three critical deficiencies that collectively prevent the deployment of resource-efficient, prescriptive systems in low-resource educational settings.

Firstly, despite the development of high-performing predictive models (e.g., Goren et al., Borges et al. [10, 11]), these systems remain fundamentally diagnostic. The platforms described excel at identifying risk but lack an embedded, mathematically rigorous mechanism to translate that score into a dynamically optimized intervention choice in real-time.

Secondly, where sequential decision-making is addressed, the core theoretical breakthrough—the Multi-Armed Bandit with Abandonment (MAB-A)—is optimized solely for minimizing theoretical regret while safeguarding against abandonment. Crucially, the MAB-A model neglects the differential cost of intervention resources. In the real-world context of underserved areas, where a mentor phone call costs significantly more than an automated SMS, any system that treats all actions equally in terms of resource expenditure is rendered economically unviable.

Thirdly, while explainable AI is recognized as necessary for teacher adoption, complex models often require rich feature sets that violate low-resource constraints. Therefore, a significant gap exists in creating an integrated, prescriptive framework that effectively combines a robust, minimal-feature prediction pipeline with a State-Aware, Cost-Sensitive Bandit (SCSB) policy.

3 Methodology

3.1 Dataset

For this study, we utilize the Open University Learning Analytics Dataset (OULAD). This dataset contains comprehensive, anonymized data from the Virtual Learning Environment (VLE) of the Open University, a large distance-learning university in the United Kingdom. It is a well-established benchmark for research in student dropout prediction and learning analytics due to its size, detail, and real-world origin. The dataset encompasses student demographic information, registration details, and fine-grained daily interaction logs with online learning materials.

Dataset Details:

- **Size:** Records of 32,593 students and over 10.6 million VLE interaction rows.
- **Attributes:** Includes `studentInfo.csv` (gender, region, education), `studentRegistration.csv` (dates), and `studentVle.csv` (clicks, interactions).
- **Labeling:** The `final_result` column serves as the ground-truth label.

3.2 Data Challenges

The OULAD dataset, while comprehensive, presents several challenges that must be addressed:

- **Class Imbalance:** The "Withdrawn" category is significantly smaller than the combined "Pass," "Fail," and "Distinction" categories, requiring specialized techniques to prevent model bias.
- **Data Integration:** Information is distributed across seven distinct files. A significant challenge is to correctly merge these tables to construct a unified feature set for each student.
- **Temporal Feature Engineering:** Raw clickstream data must be aggregated to engineer features that capture a student's engagement trajectory.

3.3 Feature Engineering Techniques

Feature engineering involves creating derived metrics like cumulative grades and engagement scores. Formulations:

- **Cumulative Grade:**

$$N_{\text{cumulative}} = \sum_{i=1}^{t} N_i \tag{1}$$

- **Grade Difference:**

$$\Delta N = N_t - N_{t-1} \tag{2}$$

- **Z-Score Normalization:**

$$z = \frac{\text{value} - \mu_{\text{course}}}{\sigma_{\text{course}}} \tag{3}$$

These features are core to achieving interpretability and creating models with minimal data dependency.

3.4 Synthetic Minority Over-sampling Technique (SMOTE)

SMOTE is a data augmentation technique designed to address class imbalance. It generates synthetic samples for the minority class by interpolating between existing minority instances. New samples are generated via:

$$x_{\text{new}} = x_i + \lambda \left(x_{\text{neighbor}} - x_i \right), \quad \lambda \in [0, 1] \tag{4}$$

This technique is crucial for improving Recall in imbalanced classification problems like dropout prediction.

3.5 Neural Network for Prediction

The prediction component utilizes a Feed-Forward Neural Network (FFNN) designed to estimate the probability of student attrition based on the engineered features. The model architecture consists of an input layer corresponding to the feature vector size, multiple hidden layers with ReLU activation functions to capture non-linear dependencies, and a final output layer with a sigmoid activation function to output a probability score between 0 and 1.

The FFNN is trained using binary cross-entropy loss, optimizing the weights to minimize the difference between the predicted probability and the actual dropout label. Regularization techniques such as dropout layers are employed to prevent overfitting.

3.6 Restless Multi-Armed Bandits (RMAB) for Intervention

The intervention strategy is modeled using the Restless Multi-Armed Bandit (RMAB) framework. Unlike standard bandits where the state of an arm only changes when it is pulled, in RMAB, the state of each student (arm) evolves over time regardless of the action taken.

The goal is to select a subset of students for intervention at each time step to maximize the total expected engagement (or minimize dropout) subject to a budget constraint. The "arms" represent the students, and the "actions" represent the different types of interventions (e.g., SMS, email, mentor call). The effectiveness of an intervention depends on the student's current state (e.g., engaged or at-risk).

4 Proposed Architecture

4.1 Overview

The proposed system combines a neural churn-prediction model with a budget-constrained adaptive intervention engine based on multi-armed bandits. The complete pipeline is organized as a two-loop learning system.

The architecture is organized into two complementary feedback loops that operate at different time scales:

- A **slow learning loop**, in which a neural network is periodically retrained on historical student data to predict dropout risk at the level of an academic term.
- A **fast learning loop**, in which a bandit agent continuously updates the estimated effectiveness of different interventions from real-time feedback while operating under a fixed intervention budget (e.g., weekly).

A teacher-facing dashboard sits on top of these loops and keeps a human in the loop for transparency, interpretability, and oversight.

4.2 Data Ingestion and Preprocessing

Raw student records are first passed through a Data Ingestion and Preprocessing layer. This component performs cleaning, type-casting, missing-value imputation, and feature normalization. The output is a matrix of clean student vectors used by both the prediction and the bandit modules.

4.3 Slow Loop: Churn Prediction Engine

The Churn Prediction Engine is implemented as a feed-forward neural network trained to estimate the probability that a student will churn in the current term. The model consumes the preprocessed features and outputs a risk score $p_\theta(y = 1 \mid x)$ for each student.

At the end of every academic period, the model is updated in a Model Retraining step using the accumulated historical data. The output of this loop is a prioritized student list sorted by predicted risk.

4.4 Fast Loop: Adaptive Intervention Engine

4.4.1 Problem Formulation: Restless Multi-Armed Bandits (RMAB)

The Adaptive Intervention Engine formulates the resource allocation problem as a Restless Multi-Armed Bandit (RMAB). In this framework, each student is modeled as an arm whose state (engaged vs. at-risk) evolves over time, and the goal is to apply interventions to maximize the total number of engaged students.

4.4.2 Definition of Arms (Interventions)

We define the set of available actions (Arms), denoted as $\mathcal{A} = \{a_0, a_1, a_2, a_3\}$, each representing a specific intervention strategy with an associated cost $C(a_k)$:

1. **No Action (a_0):** The baseline state where no intervention is applied. Cost $C(a_0) \approx 0$.
2. **SMS Reminder (a_1):** A low-cost, automated nudge sent to the student's mobile device. This is suitable for minor engagement lapses.
3. **Practice Quiz / Additional Assignment (a_2):** A medium-cost automated content push designed to re-engage the student academically.
4. **Mentor Call (a_3):** A high-cost, one-to-one human interaction. This is the most effective but most expensive resource.

4.4.3 Cost-Sensitive Decision Logic

Standard bandit algorithms maximize purely the probability of reward (e.g., probability of student retention). However, in resource-constrained settings, we must balance the efficacy of an intervention against its cost.

We define $\Delta\text{Risk}(a_k)$ as the **reduced risk** (or increase in retention probability) achieved by applying intervention a_k. This is estimated dynamically by the algorithm. To make the best decision, we utilize a **Net Utility Score** that explicitly calculates the cost-adjusted risk reduction (represented here via a Lagrange multiplier formulation).

Let θ_k be the estimated probability of success (Retention) for intervention a_k. The utility score U_k is calculated as:

$$U(a_k) = \text{Reduced Risk} - \lambda \cdot \text{Cost} = \theta_k - \lambda \cdot C(a_k) \quad (5)$$

Where:

- $\theta_k \in [0, 1]$ is the estimated reduction in churn risk (i.e., probability of staying) provided by the algorithm's belief state.
- $C(a_k)$ is the normalized cost of the chosen intervention.
- λ is a tunable parameter representing the "price of risk." It determines how much cost the system is willing to incur for a unit decrease in churn risk.

The decision rule selects the arm that provides the maximum benefit after accounting for the cost penalty:

$$a^* = \arg\max_k \left(\theta_k - \lambda\, C(a_k)\right) \quad (6)$$

4.4.4 Optimization Algorithms

We implement and compare two learning strategies to solve this allocation problem:

1. ϵ-Greedy Algorithm. The ϵ-greedy learner explores each arm with probability ϵ and otherwise exploits the current best arm.

$$a_t = \begin{cases} \text{random arm from } \mathcal{A} & \text{with probability } \epsilon \\ \arg\max_{a \in \mathcal{A}} \hat{Q}(a) & \text{with probability } 1 - \epsilon \end{cases} \quad (7)$$

where $\hat{Q}(a)$ is the average reward observed for arm a so far.

2. Bayesian Thompson Sampling (TS). We propose a Bayesian Thompson Sampling learner that maintains a probabilistic belief about the efficacy of each intervention. Since the outcome Y is binary (Stay/Churn), we model the reward distribution using a Beta-Bernoulli conjugate prior.

For each arm k, the system maintains two parameters:

- α_k: The count of successes (students retained) when arm k was used.
- β_k: The count of failures (students churned) when arm k was used.

Algorithm Steps:

1. **Sampling:** At each round t, for each arm k, sample a value θ_k from the Beta distribution:
$$\theta_k \sim \text{Beta}(\alpha_k,\, \beta_k) \quad (8)$$
2. **Cost Adjustment:** Calculate the Net Utility Score using the risk reduction/cost formula:
$$\text{Score}_k = \theta_k - \lambda \cdot C(a_k) \quad (9)$$
3. **Selection:** Select the arm with the highest Score: $a^* = \arg\max_k(\text{Score}_k)$.
4. **Update:** After observing the actual outcome $y_t \in \{0, 1\}$, update the parameters for the chosen arm:
$$\alpha_{a^*} \leftarrow \alpha_{a^*} + y_t \quad (10)$$
$$\beta_{a^*} \leftarrow \beta_{a^*} + (1 - y_t) \quad (11)$$

This approach allows the system to balance exploration and exploitation naturally. Arms with high uncertainty (low sample counts) will have wider Beta distributions, encouraging exploration. As data accumulates, the distributions narrow, and the system exploits the most cost-effective interventions.

4.5 Teacher Dashboard and Human Feedback

The Teacher Dashboard presents, for each round, the list of students, their model-predicted risk scores, and the recommended interventions. The teacher may accept, reject, or modify the proposed plan. The executed plan and the realized student outcomes are then fed back to both the neural network retraining module (slow loop) and the bandit reward-update module (fast loop). This design keeps a human in the loop, improving trust in the system.

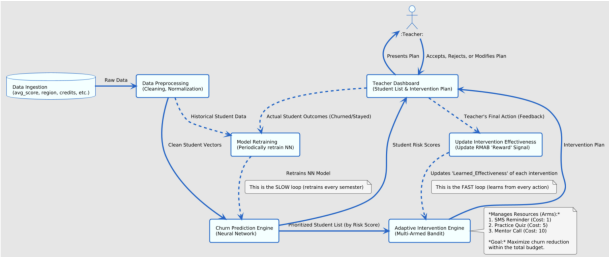


Figure 1: Proposed system architecture. The slow loop retrains the neural risk model, while the fast loop uses a Restless Multi-Armed Bandit to adapt interventions.

5 Experimental Results

This section evaluates the performance of the proposed two-stage system consisting of (1) a neural churn-prediction model and (2) a Restless Multi-Armed Bandit (RMAB) intervention engine. Experiments were conducted on processed student data with synthetically simulated intervention feedback.

5.1 Churn Prediction Model Performance

The neural network classifier was trained for 70 epochs using the Adam optimizer. The training and validation loss curves exhibited stable convergence (Figure 2).



Figure 2: Training and validation loss across epochs.

The final predictive performance on the test set is summarized in Table 1.

Table 1: Neural network performance metrics

Metric	Score
Accuracy	0.731
Precision	0.774
Recall	0.726
F1-score	0.749
ROC-AUC	0.808
SMAPE	11.37

Feature importance analysis using XGBoost revealed that `avg_score`, `num_of_prev_attempts`, `highest_education`, and `studied_credits` were the strongest predictors of churn (Figure 3).

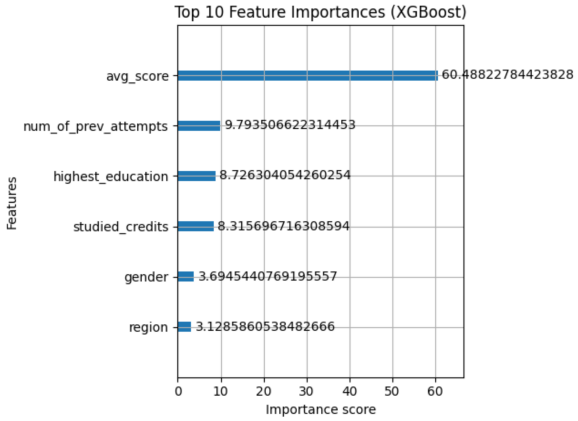


Figure 3: Top feature importances derived using XGBoost.

5.2 RMAB Simulation and Churn Reduction

5.2.1 Reduction in Expected Churn Rate

The RMAB simulation was run for 10 rounds. The expected churn rate dropped steadily from approximately 0.54 to about 0.51 by Round 10 while remaining within the fixed intervention budget (Figure 4).

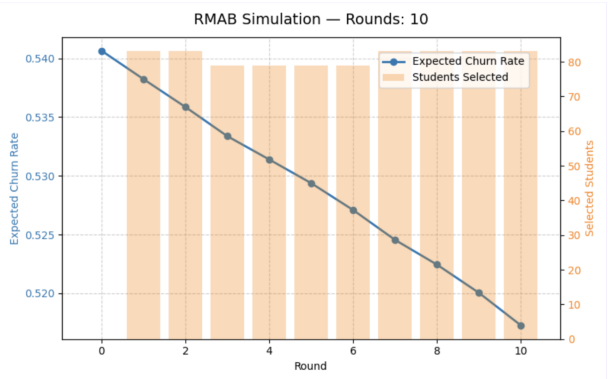


Figure 4: RMAB simulation: expected churn rate and students selected per round.

5.2.2 Success Rates and Action Accuracy

We compared the Thompson Sampling (TS) policy against the ϵ-greedy baseline.

Thompson Sampling Performance:

- Average success rate: 0.108
- Average action accuracy: 0.95
- Cumulative regret: 14
- Policy efficiency: 0.986

ϵ-Greedy Performance:

- Cumulative regret: 45
- Action accuracy: 0.876
- Policy efficiency: 0.918

Thompson Sampling consistently achieved lower cumulative regret and higher policy efficiency compared to the ϵ-greedy approach.

5.2.3 Belief Convergence

The evolution of learned arm efficacies demonstrates the effectiveness of the Bayesian update rule. Thompson Sampling quickly converged to the true synthetic efficacies (SMS $=$ 0.10, Assignment $=$ 0.25, Mentor Call $=$ 0.50). This rapid convergence minimizes the cost of exploration and maximizes the impact of the interventions.

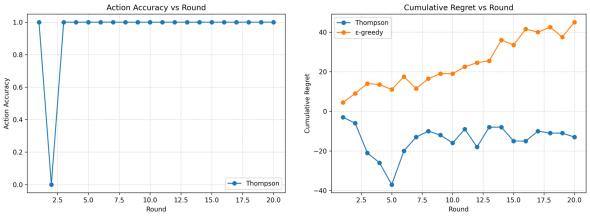


Figure 5: Comparison of Thompson Sampling vs. ϵ-greedy: (a) Action Accuracy, (b) Cumulative Regret.

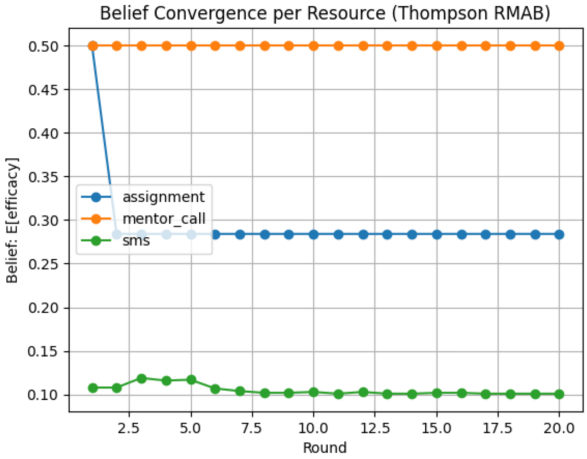


Figure 6: Convergence of Thompson Sampling beliefs to true arm efficacies.

6 Conclusion and Future Scope

This work presented a two-stage decision-support framework for reducing student dropout in technology-enhanced learning environments. The first stage uses a neural churn-prediction model to estimate student-level risk scores from routinely collected features. The second stage employs a budget-constrained Restless Multi-Armed Bandit to select personalized interventions (SMS reminder, additional assignment, mentor call) for the highest-risk students within a fixed budget.

The neural network achieved an accuracy of 0.731 and ROC-AUC of 0.808, demonstrating that the available administrative features are sufficiently predictive for risk stratification. Feature importance analysis showed that

academic performance and study history are the dominant determinants of churn.

On top of these predictions, the RMAB engine learned to allocate interventions efficiently. In simulation, the expected churn rate decreased monotonically across rounds while staying within the available budget. Thompson Sampling consistently outperformed the ϵ-greedy baseline, achieving lower cumulative regret, higher policy efficiency, and near-perfect action accuracy once its beliefs converged to the true arm efficacies. These results suggest that a Bayesian bandit formulation is well suited for adaptive intervention planning in educational settings where intervention costs are heterogeneous and feedback arrives incrementally.

Future Work. Future extensions of this work will focus on four key directions:

1. **Richer Student Representations:** Incorporating temporal activity traces, clickstream features, and textual data (forum posts, assignment feedback) to capture finer-grained learning behaviors and affective states.

2. **Contextual and Causal Bandits:** Extending the RMAB module to contextual bandit or reinforcement learning formulations that condition the intervention policy on student covariates and estimate heterogeneous treatment effects.

3. **Fairness and Constraints:** Integrating fairness-aware constraints to ensure that at-risk subgroups (e.g., late joiners, part-time students) are not systematically under-served by the optimization objective.

4. **Real-World Deployment and A/B Testing:** Embedding the system within an actual institutional dashboard for teachers, performing controlled A/B experiments against existing manual practices.

Overall, the proposed architecture demonstrates that combining supervised risk prediction with adaptive, budget-aware intervention optimization is a viable path towards intelligent, data-driven student support systems.

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